

Orphan Node		Orphan Node			
user_intent		discourse_state			
information_request	opinion_expression	previous_topic	new_topic		
0.7	0.3	0.6	0.4		
User normally ask question rather than simply express opinion				conversation often starts with previous talks	
robot_intent					
		acknowledge	provide_statement		
information_request	previous_topic	0.3	0.7		
information_request	new_topic	0.4	0.6		
opinion_expression	previous_topic	0.7	0.3		
opinion_expression	new_topic	0.8	0.2		
				user_preference	
				long_statements	short_statements
				0.5	0.5
				User preference could be 50 50 for long and short statements	
robot_revised_intent					
		acknowledge_short	acknowledge_long	provide_statement_short	provide_statement_long
acknowledge	long_statements	0.2	0.4	0.1	0.3
acknowledge	short_statements	0.4	0.2	0.2	0.2
provide_statement	long_statements	0.2	0.1	0.2	0.5
provide_statement	short_statements	0.1	0.1	0.7	0.1
For direct combination like acknowledge_long for acknowledge and long statement the prob is higher in the row same for each row					
Determine the probability of robot_intent = acknowledge given the evidence user_intent = information_request and user_preference = long_statements.					
P(robot_intent=acknowledge user_intent=information_request,user_preference=long_statements)					
P(robot_intent=acknowledge =P(acknowledge information_request,previous_topic)P(previous_topic)					
+P(acknowledge information_request,new_topic)P(new_topic)					
0.3	0.6		0.4	0.4	
0.18			0.16		
		0.34			
How does the probability change when we add the evidence discourse_state = new_topic to the above? Explain your observation.					
P(robot_intent=acknowledge discourse_state=new_topic,user_preference=long_statements)					
P(robot_intent=acknowledge =P(acknowledge new_topic,information_request)P(information_request)					
+P(acknowledge new_topic,opinion_expression)P(opinion_expression)					
0.4	0.7		0.8	0.3	
0.28			0.24		
		0.52			
However , If adding discourse state to the part a gives					
it changes from 0.34 to 0.40 , as 0.40 is information request,newtopic and acknowledge					
After adding the evidence discourse_state = new_topic, the network no longer needs to consider previous_topic					
so P(acknowledge)=0.4					
that is increased by 6 percent points					